

Example Quiz (Transactions and Locks Part I)

Instructions: For each problem, you must perform all of the following tasks correctly in order to get credit.

1. Circle all SQL commands that block.
2. Write an X through all commands that error.
3. If a command causes a deadlock, write DEADLOCK and stop the problem.
4. Write the output of each SELECT statement.

You will be allowed to use a computer and access the internet, but you will not be allowed to connect to postgres and run postgres commands.

Problem 1:

Session 1

```
1 CREATE TABLE t ( a INT );
2
3 BEGIN;
4 LOCK TABLE t IN SHARE MODE;
5
6
7
8 LOCK TABLE u IN ROW EXCLUSIVE MODE;
9
10
11 COMMIT;
```

Session 2

```
1
2 CREATE TABLE u ( a INT );
3
4
5 BEGIN;
6 LOCK TABLE u IN
7 SHARE ROW EXCLUSIVE MODE;
8
9 LOCK TABLE t IN
10 ROW EXCLUSIVE MODE;
11
12 COMMIT;
```

Problem 2:

Session 1

```
1 CREATE TABLE t (  
2     a INT PRIMARY KEY  
3 );  
4 BEGIN ISOLATION LEVEL  
5 REPEATABLE READ;  
6 INSERT INTO t VALUES (5);  
7  
8  
9 INSERT INTO t VALUES (6);  
10 SELECT count(*) FROM t;  
11  
12  
13 COMMIT;
```

Session 2

```
1  
2  
3  
4  
5  
6  
7 BEGIN;  
8 INSERT INTO t VALUES (6);  
9  
10  
11 INSERT INTO t VALUES (5);  
12 ABORT;
```

Problem 3:

Session 1

```
1 CREATE TABLE t ( a INT );  
2 INSERT INTO t VALUES (9);  
3 INSERT INTO t VALUES (10);  
4 BEGIN;  
5  
6 UPDATE t SET a = a+1;  
7  
8 COMMIT;
```

Session 2

```
1  
2  
3  
4  
5 BEGIN;  
6  
7 DELETE FROM t WHERE a=10;  
8  
9 SELECT count(*) FROM t;  
10 COMMIT;
```

Problem 4:

Session 1

```
1 CREATE TABLE t ( a INT );
2 INSERT INTO t VALUES (1);
3 INSERT INTO t VALUES (2);
4 INSERT INTO t VALUES (3);
5
6
7
8 BEGIN;
9 UPDATE t SET a=6 WHERE a=1;
10
11
12 INSERT INTO t VALUES (1);
13 COMMIT;
```

Session 2

```
1
2
3
4
5 BEGIN ISOLATION LEVEL
6 REPEATABLE READ;
7 SELECT count(*);
8
9
10 DELETE FROM t;
11 COMMIT;
12
13
14 SELECT count(*) FROM t WHERE a=1;
```